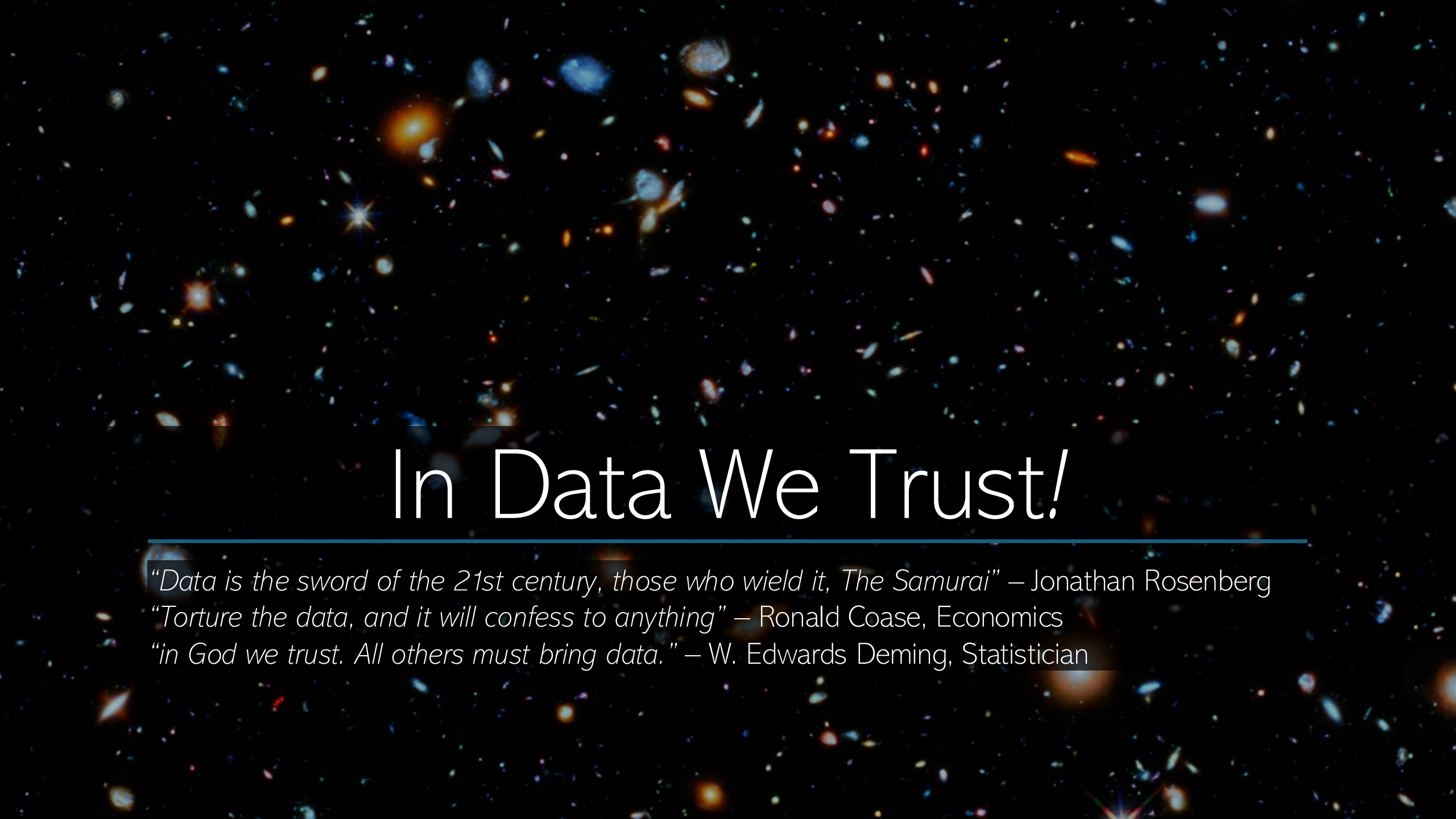


The background of the slide is a high-resolution astronomical image, likely from the Hubble Space Telescope, showing a dense field of galaxies. The galaxies are in various stages of evolution, some appearing as bright, compact cores, others as diffuse, irregular shapes. The colors range from deep blues and purples to bright oranges and yellows, representing different wavelengths of light. The overall effect is a sense of immense scale and cosmic diversity.

In Data We Trust!

"Data is the sword of the 21st century, those who wield it, The Samurai" – Jonathan Rosenberg

"Torture the data, and it will confess to anything" – Ronald Coase, Economics

"in God we trust. All others must bring data." – W. Edwards Deming, Statistician

2

Outline: Week 02 A: Monday, B: Wednesday, C: Thursday

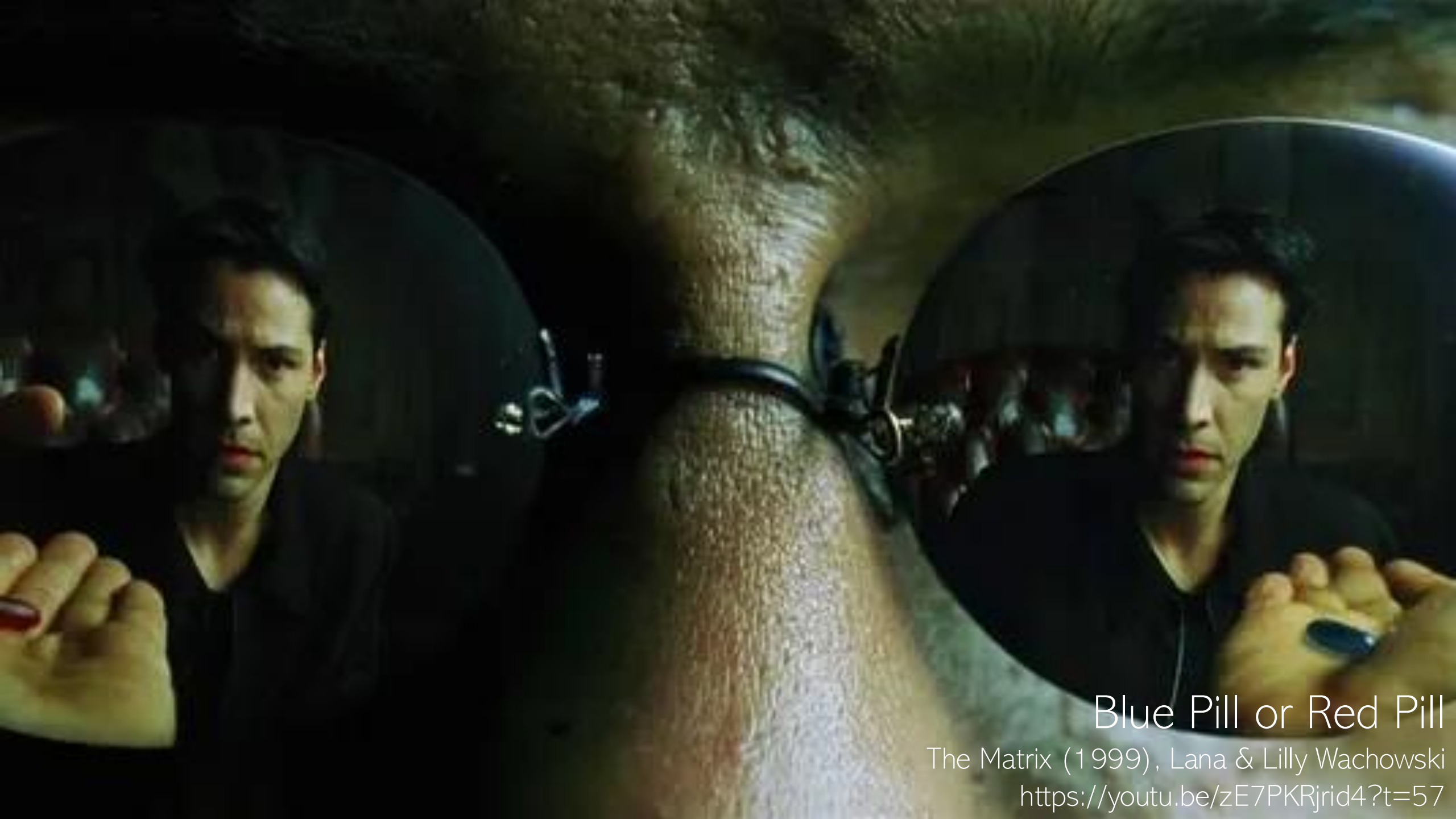
I) Data Mining (Review)

II) Course Logistics

III) Brightspace Tour

III) Another Brick in the Wall

Outline: Week 02 B: Wednesday, Topics Starts



Blue Pill or Red Pill

The Matrix (1999), Lana & Lilly Wachowski

<https://youtu.be/zE7PKRjrid4?t=57>

Grand Défi (Big Challenge) , Bronze, 175x145x130cm, Nicolas LAVARENNE
<https://www.artseiller.com/en/nicolas-lavarenne-en>



Follow the white rabbit.

Follow the White Rabbit
- The Matrix (1999), Lana & Lilly Wachowski

6

Artificial Intelligence: Let's develop one ...

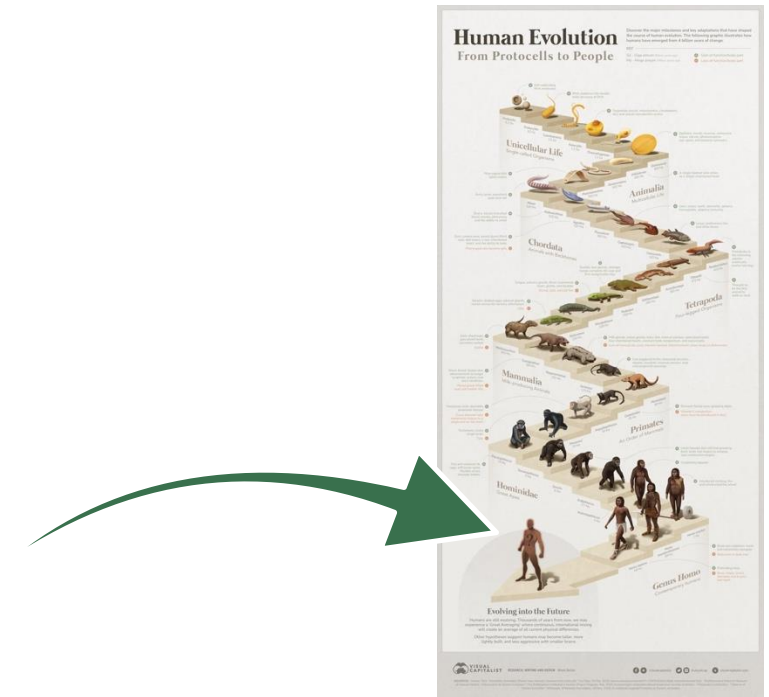
How to emulate human behaviour/actions/... ?

How human obtain behaviour/actions/...?

If I follow the same way that human obtain behaviour/actions/...

Am I going to expose same or similar behaviour/actions/ ...

Am I going to be intelligent?



7

Artificial Intelligence: Simplify more ...

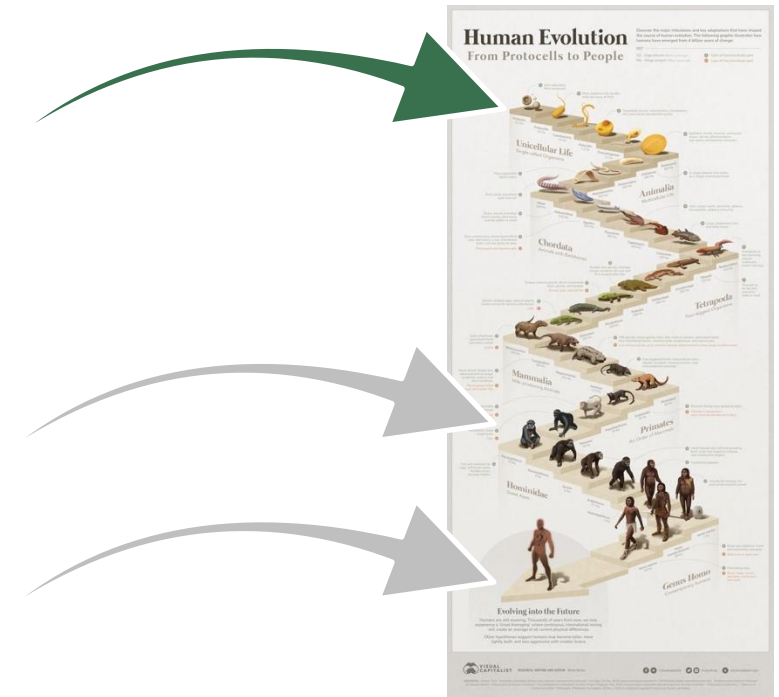
How to emulate **bacteria** behaviour/actions/... **with no complexity?**

How bacteria obtain behaviour/actions/...?

If I follow the same way that bacteria obtain behaviour/actions/...

Am I going to expose same or similar behaviour/actions/ ...

Am I going to be intelligent?



8

Artificial Intelligence

Machines emulate bacteria learning/training to obtain bacteria-level intelligence

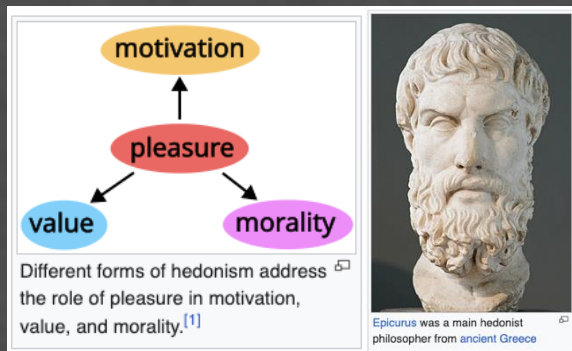
Bacteria don't have brains, but they can:

Run: Move in a direction **for food (nutrients)**

Tumble: Randomly change direction **when fewer nutrients or hitting a barrier**

Are they intelligent, or not intelligent (crazy)?

They want to survive, live longer, grow community, enjoy ...



Hedonism is a family of philosophical views that prioritize pleasure.

1 μm

9

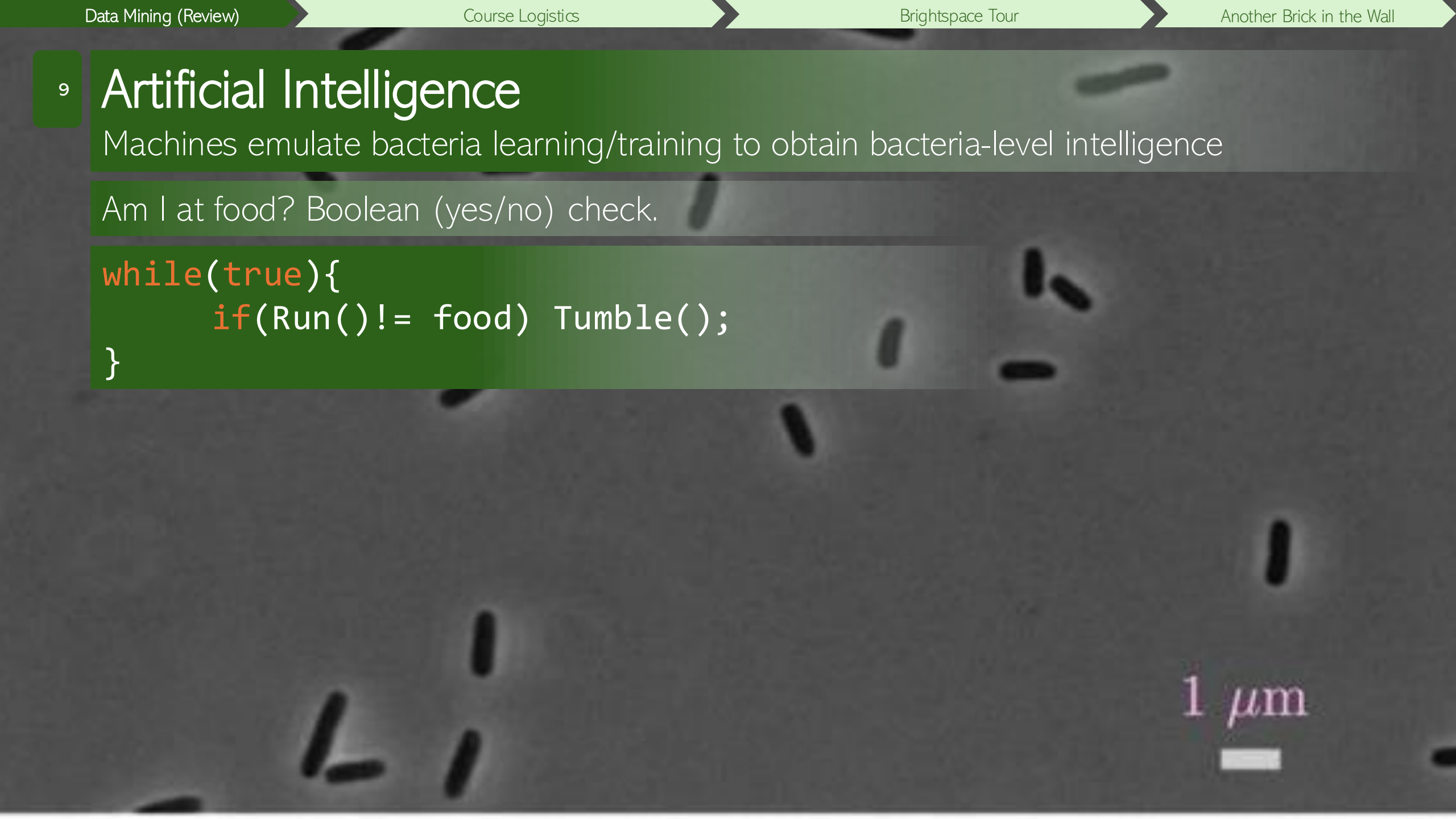
Artificial Intelligence

Machines emulate bacteria learning/training to obtain bacteria-level intelligence

Am I at food? Boolean (yes/no) check.

```
while(true){  
    if(Run() != food) Tumble();  
}
```

1 μm

A grayscale microscopic image showing numerous rod-shaped bacteria. In the bottom right corner, there is a white horizontal scale bar with the text "1 μm" above it.

10

Artificial Intelligence: Algorithm (Machine) Learning

Machines emulate bacteria learning/training to obtain bacteria-level intelligence

2026: <https://www.newscientist.com/article/2519284-single-celled-organism-with-no-brain-capable-of-pavlovian-learning/>

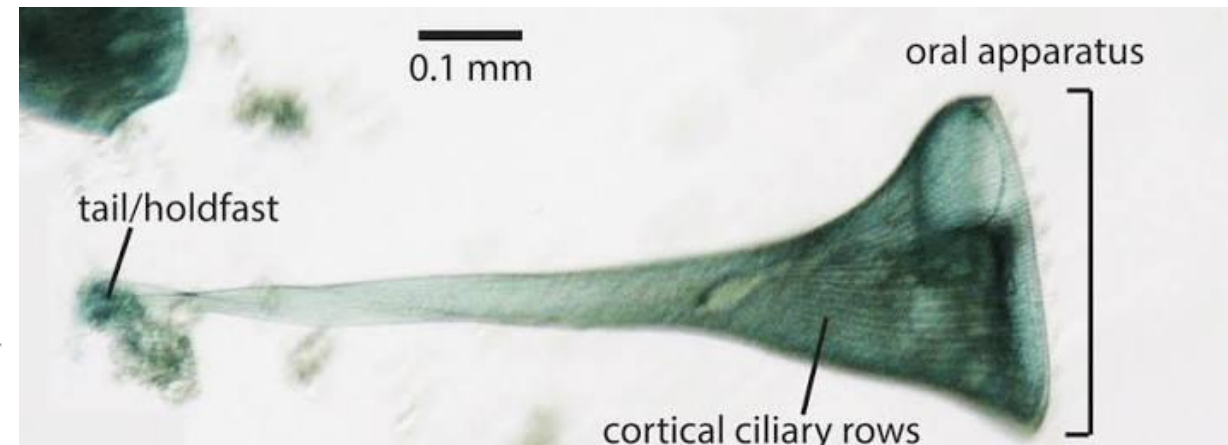
What if we add a single element of brain/memory/bit

To record a single history ...

To record a single experience ...

"been there, done that" → Benefit from experience!

Then, the researchers did two more set-ups: A weak-strong tap sequence (separated by one second) and a weak-weak tap sequence. They found that protists in the first group displayed an enhanced reaction to the first weak tap, as if they perceived that it predicted an accompanying strong tap, until their responses eventually decreased over time as predictive habituation took over. The weak-weak group showed no such enhanced reaction, suggesting that *S. coeruleus* really could learn to associate a weak tap with a forthcoming strong tap.



Stentor Coeruleus STEN-ter see-ROO-lee-us / 'stɛntər sɪ'ruːliəs/

11

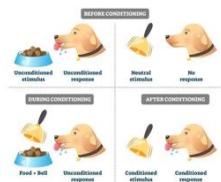
Artificial Intelligence: Algorithm (Machine) Learning

Machines emulate bacteria learning/training to obtain bacteria-level intelligence

2026: <https://www.newscientist.com/article/2519284-single-celled-organism-with-no-brain-is-capable-of-pavlovian-learning/>

Food $\rightarrow$ Lower $P \rightarrow$ Less Tumbling $\rightarrow$ Keep Running.

```
P = 1; //prob of tumbling
while (true) {
    if (Run() == food) P-=0.1;
    if (random() < P) Tumble();//uniform random number in [0, 1]
}
```



Pavlovian Learning: Pavlov's dog Experiment: https://en.wikipedia.org/wiki/Classical_conditioning

The famous experiments by Russian physiologist Ivan Pavlov that led to the discovery of classical conditioning, respondent conditioning, or pavlovian conditioning; A behavioral procedure in which a biologically potent stimulus (e.g. food, a puff of air on the eye) is paired with a neutral stimulus (e.g. the sound of a musical triangle).

12

Artificial Intelligence: Algorithm (Machine) Learning

Machines emulate **human** learning/training to obtain **human**-level intelligence

What if we add many elements of brain/memory/bit → **Trillions** of bytes (petabytes for human)

To record history ...

To record experience ...

"been there, done that" → Benefit from **many** experiences!

→ Learning from recorded experiences (data)!

→ Data Mining

13

Artificial Intelligence: Algorithm (Machine) Learning

Machines emulate human learning/training to obtain human-level intelligence

*how does it feel,
how does it feel to be without a home,
like a complete unknown,
like a* 

*apple: no English grammar experience
table: no semantic experience
clown: no discourse (poem) experience
rolling stone: no pragmatic (why) experience*

14

Artificial Intelligence: Algorithm (Machine) Learning

Machines emulate human learning/training to obtain human-level intelligence

*how does it feel,
how does it feel to be without a home,
like a complete unknown,
like a **rolling stone** — **Bob Dylan***

<https://songmeanings.com/songs/view/5730/>



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16

Data Mining vs. Structural Search

Word Segmentation (Tokenization)

Word boundary: whitespace

Natural word delimiter

`split()`

Exceptions

New York, rock 'n' roll

Contractions: I'm

Japanese, Chinese, Thai don't have spaces between word

Hashtags: #nlproc.

Learning to Tokenize ...

Byte-Pair Encoding (BPE): Sennrich, et al., (2016). Neural machine translation of rare words with subword un
 Wordpiece: Wu et al. (2016) Google's neural machine translation system: Bridging the gap between human ar
 MaxMatch in BERT: Devlin et al. (2019). BERT: Pretraining of deep bidirectional transformers for language unc
 SentencePiece: Kudo, T. and Richardson, J. (2018). SentencePiece: A simple and language independent subwo



Outline: W02A

I) Data Mining (Review)

II) Course Logistics

III) Brightspace Tour

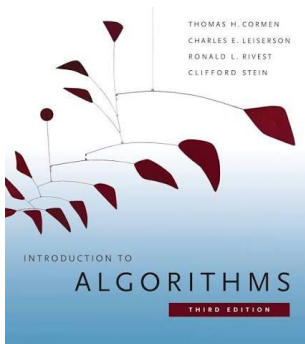
III) Another Brick in the Wall

Background

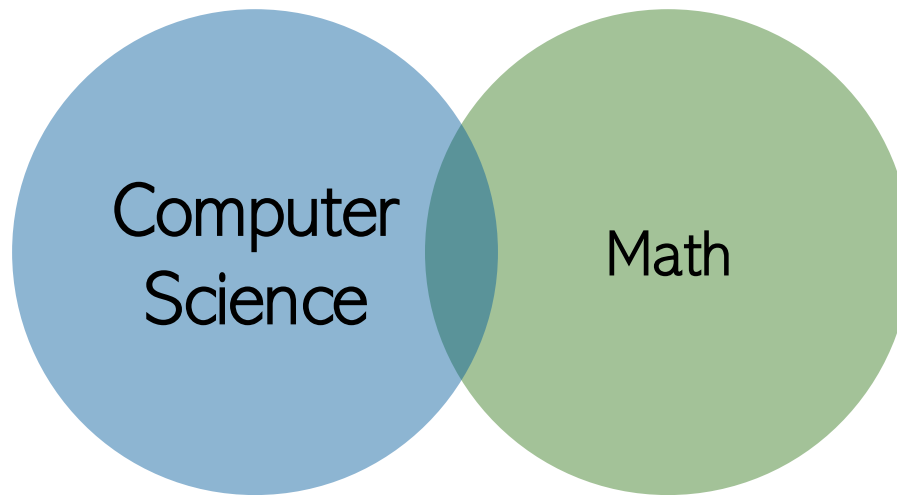
Design & Analysis of Algorithms

Greedy
Dynamic Programming
Divide-Conquer
Recursion
Backtracking
Big O

Data Structure & Modeling



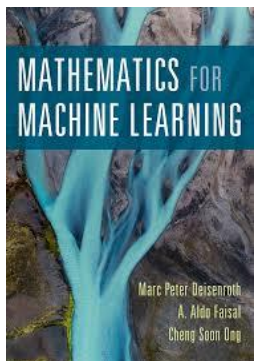
Introduction to Algorithms
By Cormen et al., 2009.



Multivariate Calculus

Derivatives
Partial Derivatives
Linear Algebra
Vectors
Matrices
Probability & Statistics

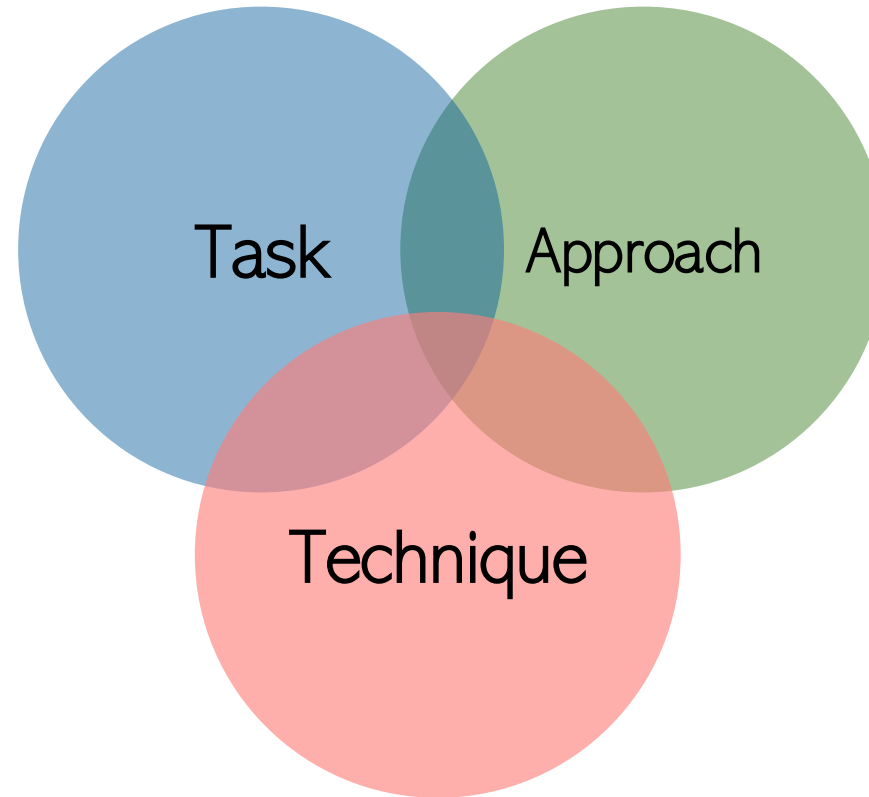
Mathematics for Machine Learning,
Deisenroth et al., 2020, <https://mml-book.github.io/>



19

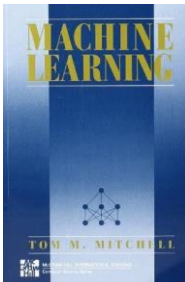
Topics

Classification
Regression
Clustering
Ranking (Recommendation)
Vector Representation and
Dimensionality Reduction
Language Modeling



Full/Self/Semi-Supervised Learning
Unsupervised Learning
Reinforcement Learning

Naive Bayes, Logistic Regression
Artificial Neural Networks, Graph Neural Network
Support Vector Machines
Principal Component Analysis
K-Nearest Neighbor, K-Means



Machine Learning, Thomas Mitchell, 1997

Role

Data Mining Designer

Data Mining Developer

Data Mining Professional User

Data Mining End User

Who designs a new learning algorithm

Who applies existing learning algorithm

Who knows how learning algorithm works

Who uses products of learning algorithms

21

Assessment

UG: Undergrad, GR: Grad

Team Project
UG: 50%
GR: 60%

Midterm Exam 1 (Week 07, Oct. 19)
10%

Midterm Exam (Week 10, Nov. 12)
10%

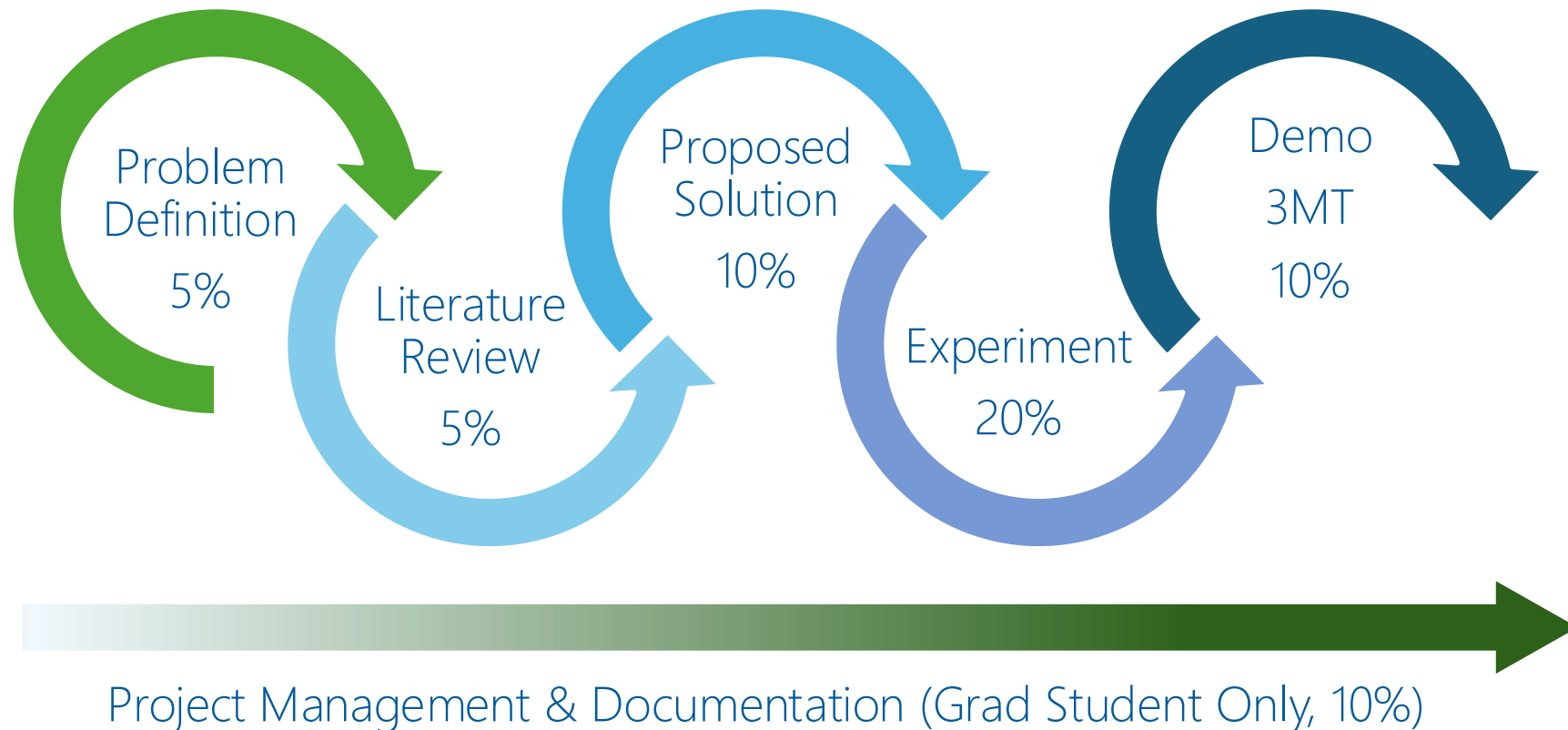
Final Exam (Week 15)
UG: 30%
GR: 20%

22

Team Project

Min. 2, Max. 3 UG Members + 1 GR

Course-long Project vs. Final Project



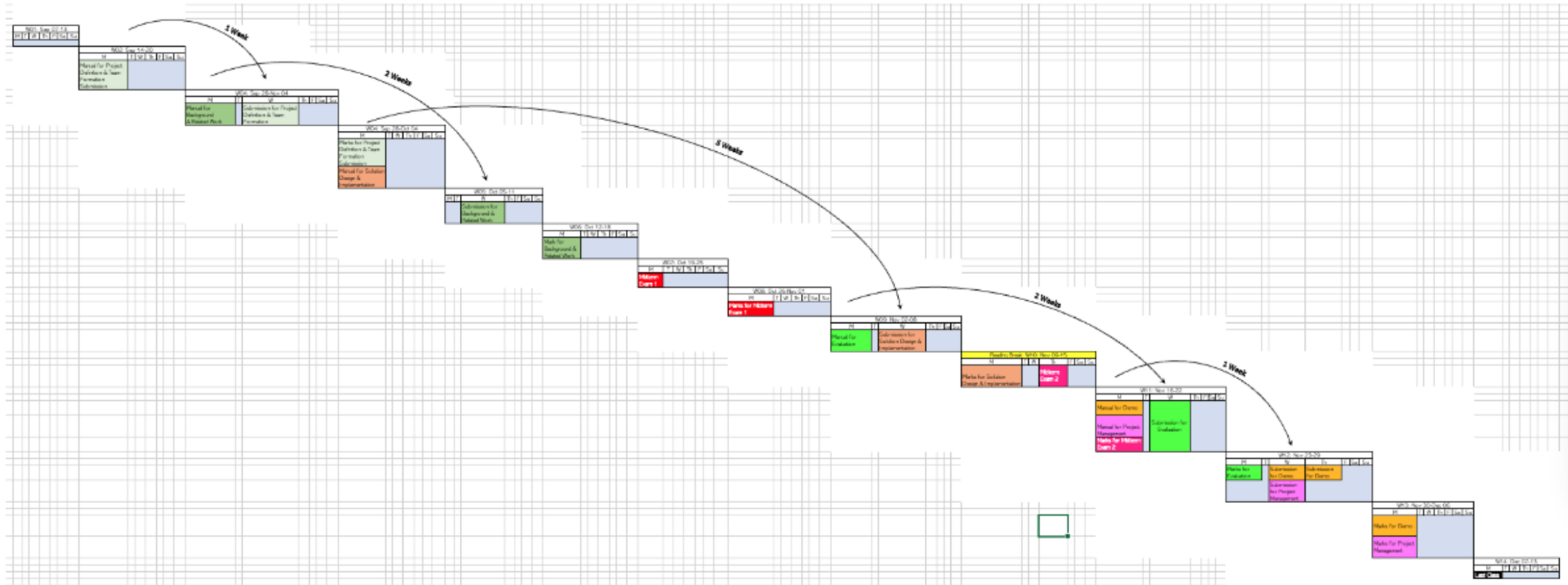
23

Weekly Schedule

MON	TUE	WED	THU	FRI	SAT	SUN
New Submission Manual Prev. Submission Marks Release		Submission Deadline Midnight				
	B02: 10:30 - 11:20 Engineering Lab Wing B215: Reia	B03: 10:30 - 11:20 Engineering Lab Wing B215: Reia				
		B04: 11:30 - 12:20 Engineering Lab Wing B215: Ali				
		B05: 13:30 - 14:20 Engineering Lab Wing B215: Ali				
Lecture A: 2:30-3:30 Strong Building C103: Hossein		Lecture B: 2:30-3:30 Strong Building C103: Hossein	Lecture C: 2:30-3:30 Strong Building C103: Hossein			
Office A: 3:30-4:30 ECS613: Hossein		Office B: 3:30-4:30 ECS613: Hossein	Office C: 3:30-4:30 ECS613: Hossein			
Marking						
Monitor Discussion Board						
Link Lecture Video		Link Lecture Video	Link Lecture Video			

24

Timeline





If I was informed correctly, the clock is ticking.
Is that right, Jimmie?

27

Team Project: Take a Small but Novel Step!



Osprey: Enhancing Online Grooming Detection via Backtranslation Augmentation

<https://github.com/fani-lab/Osprey>

<https://doi.org/10.1145/3627673.3679974>

<https://aclanthology.org/2025.coling-main.160/>

Document Classification

Conversation Classification

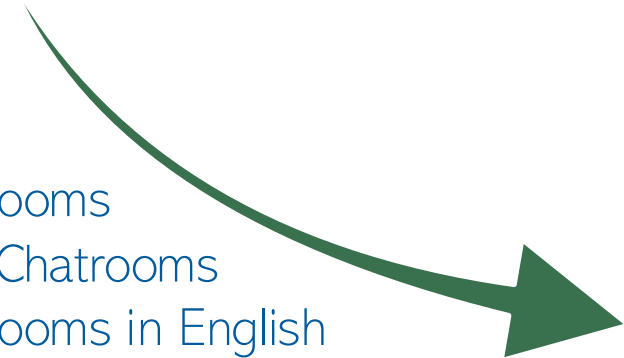
Conversation Classification in Chatrooms

Predatory Conversation Classification in Chatrooms

Predatory Conversation Classification in Chatrooms in English

The Effect of Augmentation on Predatory Conversation Classification in Chatrooms in English

The Effect of Backtranslation Augmentation on Predatory Conversation Classification in Chatrooms in English



28 Team Project: GitHub Repo

Branding: Pick a name for your project, design a logo, ...

Structured: Directories, Layers, Output, ...

Contributions: Pull Requests, Commits, ...

license.txt

readme.md: Explanation, setup, samples, ...

Scripts: For sample run, explaining architecture, ...

Scripts: **Should not be the main codebase!**

29

Team Project: GitHub Repo

Let's have a look at some good and bad repos

https://github.com/hosseinfani/security_incident_ontology

<https://github.com/fani-lab/OpeNTF>

<https://github.com/fani-lab/LADy>

<https://github.com/fani-lab/Osprey>

